

B. Tech Degree IV Semester Examination, April 2008

ME 403 APPLIED THERMODYNAMICS

(2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) State and explain second law of thermodynamics with illustrations. (10)
 (b) A fluid undergoes a reversible adiabatic compression from 0.5 MPa, 0.2 m³ to 0.05 m³ according to the law, $PV^{1.3} = \text{constant}$. Determine the change in enthalpy, internal energy and entropy and the heat transfer and work transfer during the process. (10)

OR

- II. (a) Explain why the specific heat of a saturated vapour may be negative. (10)
 (b) Derive the third T ds equation

$$Tds = C_v \left(\frac{\partial T}{\partial P} \right)_g dp + C_p \left(\frac{\partial T}{\partial V} \right)_p dv. \quad (10)$$

- III. (a) Explain the significance of Gibb's and Helmholtz functions. (5)
 (b) State Dalton's law and Amagat's law of mixture of perfect gases. (5)
 (c) Find the increase in entropy when 2 Kg of Oxygen at 60°C are mixed with 6 Kg of nitrogen at the same temperature. The initial pressure of each constituent is 103 kPa and is the same as that of the mixture. (10)

OR

- IV. (a) State and derive Gibbs-Dalton law for mixtures of ideal gases. (10)
 (b) A mixture of ideal gases consist of 3 Kg of nitrogen and 5 Kg of Carbon dioxide at a pressure of 300 kPa and a temperature of 20°C. Find (i) the mole fraction of each constituent (ii) the equivalent molecular weight of the mixture (iii) the equivalent gas constant of the mixture (iv) the partial pressures and partial volumes. (10)

- V. (a) Sketch and explain the working of a Bomb calorimeter. (10)
 (b) The product of combustion of a hydrocarbon fuel of unknown composition have the following composition as measured on a dry basis :
 $\text{CO}_2 = 8.0\% \quad \text{CO} = 0.9\% \quad \text{O}_2 = 8.8\% \quad \text{N}_2 = 82.3\%$
 Calculate the air fuel ratio. (10)

OR

- VI. (a) Write short notes on the following : (10)
 (i) Excess air
 (ii) Adiabatic flame temperature
 (iii) Higher and lower heating values.
 (b) The lower heating value of propane at constant pressure and 25°C is 20,440.09 KJ/Kg mole. Find the higher heating value at constant pressure and at constant volume. (10)

(Turn Over)

- VII. (a) What is the importance of specific fuel consumption. (5)
 (b) Schematically explain the use of the study of the heat balance of an engine. (5)
 (c) Explain the effect of the following factors on the performance of an engine :
 (i) Compression ratio (ii) Air-fuel ratio
 (iii) Spark timing (iv) Engine speed
 (v) Mass of inducted charge (v) Heat loss. (10)

OR

- VIII. (a) In a test of a four-cylinder, four-stroke petrol engine of 75 mm bore and 100 mm stroke, the following results were obtained at full throttle at a constant speed and with a fixed setting of the fuel supply of 0.082 Kg/minute.
 bp with all cylinders working = 15.24 KW
 bp with cylinder number 1 cut-off = 10.45 KW
 bp with cylinder number 2 cut-off = 10.38 KW
 bp with cylinder number 3 cut-off = 10.23 KW
 bp with cylinder number 4 cut-off = 10.45 KW
 Estimate the indicated power of the engine under these conditions. If the calorific value of the fuel is 44 MJ/Kg, find the indicated thermal efficiency of the engine. Compare this with the air-standard efficiency, the clearance volume of one-cylinder being 115 CC. (12)
 (b) Define air-fuel ratio and briefly state its effect on power output, fuel consumption and combustion pressure. (8)

- IX. Write short notes on the following :
 (i) Cetane number (ii) HUCR
 (iii) Octane number (iv) Wankel engine
 (v) Turbo charging. (20)

OR

- X. (a) With suitable diagrams explain the phenomenon of Combustion in SI engines. (10)
 (b) Sketch and explain the different Combustion chambers of CI engines. (10)

